Year 12 Admission Assessment v3.1: Optional sciences, mark scheme

Version 3.1, October 2026. Use with `Y12/Y12_Optional_Sciences_v3.1` (the student paper: Biology, Chemistry and Physics, 15 marks and 15 minutes each). The core assessment (English, maths, core science, cognitive ability screen and interview) is in `Y12_Assessor_Instructions_v3.1`; this document covers the optional science papers only.

Reusable: print once and keep in the assessor's folder. Do not write on these pages. Record each candidate's marks on their **grading sheet** (`Y12_Grading_Sheet_v3.1`), which has a row for each optional science (mark / 15, band, NT, L, and a "not taken" tick).

1. What these papers are for

A Year 12 entrant joins the **first year of A Level**. The optional science papers show whether a student is ready to start an A Level course in **each science they intend to take**: they test the end of an IGCSE course (Cambridge or Edexcel style, no board named) and the ideas an A Level course assumes on day one. Each paper has 6 to 8 questions, from easy to hard, every one with a diagram, table or graph, and each includes at least one calculation, one data question, one two-mark "explain" question and one working-scientifically question. The hardest question in each paper is meant to stretch a strong IGCSE student: few candidates will score 15.

Who sits them. A student from a Chinese mainstream school has usually **completed Grade 10 (高一)** and has not sat IGCSE. Their science is often strong in content and calculation, but some topics come later in the Chinese sequence (Section 2 of each science below), and the English of science (command words, subject vocabulary, "explain" answers) is usually the gap. A student from an international or bilingual school may have taken, or be about to take, IGCSE: the papers then confirm the level.

Not standardised. The band guide (Section 7) is provisional. Review it, with the readiness guidance in Section 8, after the first 20 to 30 candidates.

2. Which sciences to give

- Agree with the family **before the day** which sciences the student intends to take at A Level, and give **only those** papers (one, two or three). Tell the student at the start which ones they will do; the cover of the paper asks them to tick them.
- A student who is **undecided** between sciences may sit the ones under consideration, up to all three. A student who will take no science sits none.
- **Order on the day:** the optional sciences come after the cognitive ability screen, each 15 minutes, with a short break after two, then the interview. If the day is too long, or the student is tired, the optional sciences (and the interview) **may be given on a second visit**: record this on the grading sheet.
- Give the sciences in any order; Biology, Chemistry, Physics as printed is the simplest. A student doing one science turns straight to that section.

3. How to give them

Paper	Time and marks	Materials	Print
Biology	15 minutes, 15 marks, 7 questions	Pencil, rubber, ruler, scientific calculator	One student paper per candidate (all three sections are in it). This mark scheme once.
Chemistry	15 minutes, 15 marks, 8 questions. Relative atomic masses are printed at the start of the section.	As above	
Physics	15 minutes, 15 marks, 7 questions. The equations are printed in a box at the start of the section.	As above	

Calculator. A **scientific calculator is allowed throughout these papers** (the paper says so at the start of each section). Give it to the student before they start. The calculations test whether the student can choose and use the right method, not arithmetic.

Access rules (the same as the rest of the pack, Section 2 of the assessor instructions):

	Optional sciences
Reading aloud	You may read **any question aloud in English**, as often as the student asks, and point to the diagram. Do not explain scientific words or hint at the method.
Chinese	If a Chinese-speaking colleague is present and the student clearly cannot follow the English after it has been read aloud, the colleague may say the question **once** in Chinese. Write **L** in the margin. The mark still counts. Record the number of L questions for each science on the grading sheet: it is evidence about English, not science.
Answers	**All answers in English**: numbers, symbols, units, words, working and diagrams all count. An answer or explanation in Chinese scores 0 (this includes the "Explain" answers).
Not yet taught (NT)	If the student says they have not learnt a topic, write **NT** beside the question. It scores 0. List the NT topics on the grading sheet; the "may not have been taught" column below says what to expect.
Extra time, a reader	Allowed for a known need: record it on the grading sheet.

Say (at the start of the first science): "You are going to do [Biology / Chemistry / Physics]. Each science has 15 minutes and 15 marks. The questions start easy and get harder. You may use your calculator. I can read any question to you in English, as often as you like, but I cannot explain the science. If you have not learnt a topic, tell me and go on to the next question. Write all your answers in English, and show your working. When you finish a science, stop at the banner and wait for me. Please turn to [Biology]. You may start."

Say (for the next science): "Thank you. Now turn to [Chemistry]. You have 15 minutes. You may start."

Timing and stopping. Keep each science to 15 minutes; give a two-minute warning. Encourage the student to look at every question, because later questions on a different topic may be easier for them. The student may stop a science when they have done what they can: say "That is fine. Thank you." If the student is upset or very tired, stop, take a break, and record it.

Marking. Use the tables below. **B1** is an independent mark for a correct answer or statement; **M1** is a method mark; **A1** is an accuracy mark that depends on the method. Where an answer is carried from an earlier part, **ecf** (error carried forward) is allowed as stated. Correct answers without working score full marks unless the table says otherwise. Units: accept the unit printed after the answer box; do not take marks off for a missing unit where one is printed. Spelling: accept any spelling of a scientific word that is clearly recognisable (for example "denaturd", "insuline"); accept the correct chemical symbol or formula in place of a name.

4. Biology (15 minutes, 15 marks)

What a Year 12 entrant may not have been taught. Chinese senior high biology begins in 高一 with cells and molecules (必修1: cell structure, enzymes, respiration, photosynthesis, osmosis), then genetics and evolution (必修2) in the second term of 高一 or in 高二. Homeostasis and the nervous and hormonal systems (选择性必修1) and ecology (选择性必修2) are **高二**. Junior high biology (初一 and 初二) covers cells, the human body, basic inheritance (dominant and recessive) and natural selection in outline, so questions 5 and 7 are usually familiar in content. Expect NT most often on question 6 (sampling with quadrats) and sometimes on question 4 (homeostasis).

Q	**Topic**	**Answer**	**Marks and partial credit**	**May not have been taught?**
1	Cell structure	C (chloroplast)	B1	Secure (初一 and 高一 必修1).

Q	Topic	Answer	Marks and partial credit	May not have been taught?
2	Enzymes and temperature (data)	(a) 40 °C (b) denatured	(a) B1: accept 39 to 41. (b) B1: denatured; "the active site has changed shape" also scores. Do not accept "killed", "destroyed", "broken", "dead" (enzymes are not alive).	Enzymes and temperature are 高一 必修1: usually taught. The English word "denatured" may be the gap: a student who writes "the enzyme's structure (shape) is changed" has the idea and scores.
3	Osmosis: data and explain	(a) 0.4 mol/dm^3 (b) Water left the potato cells by osmosis, because the solution outside was more concentrated (had a lower water potential) than the cell contents	(a) B1 (the solution in which the mass did not change). (b) B1 for water moving **out of** the cells by **osmosis**; B1 for the direction explained: from the dilute cell contents (higher water potential) to the more concentrated solution (lower water potential), or "down a water potential gradient", or across the partially permeable membrane. "Sugar moved into the potato" scores 0 for (b).	Osmosis and plant cells in solutions are 高一 必修1: usually taught. Students may use the word "water potential" or "concentration gradient": both are fine.
4	Homeostasis: blood glucose	insulin	B1. Accept recognisable spellings. "Glucagon" scores 0 (it raises blood glucose). "Adrenaline" scores 0.	Hormonal control of blood glucose is 高二 (选择性必修1); insulin is named in 初二 biology (the endocrine system). May not have been taught in full: NT is possible but the name is often known.
5	Monohybrid inheritance (Punnett square)	(a) Pp, Pp, pp (with the printed PP) (b) 25% (c) 3 : 1	(a) B1 for all three cells correct (Pp top right, Pp bottom left, pp bottom right; pP is acceptable). (b) B1: 25%; accept 1/4 or "1 in 4" written in the box. ecf from (a) is not given here: the expected answer is 25% whatever the square shows. (c) B1: 3 : 1 (purple : white). "1 : 3" scores 0; "3 : 1 : 0" scores 0.	Dominant and recessive alleles and this kind of cross are 初二 (人教版八年级下) and 必修2 (end of 高一 or 高二): usually taught. The Punnett square format itself may be new; the question shows one cell filled in.
6	Sampling with quadrats: calculation and working scientifically	(a) 6 400 (b) To avoid bias (so that the sample represents the whole field)	(a) M1 for a correct first step: mean = 40 ÷ 10 = 4 dandelions per quadrat, **or** 4 × 4 = 16 per m^2, **or** number of quadrats in the field = 400 ÷ 0.25 = 1 600; A1 for 6 400. Full marks for 6 400 with no working. Common wrong answers: 1 600 (forgot the mean, or multiplied by 1 instead of 4), 160 (forgot to convert 0.25 m^2 to 1 m^2), 40 × 400 = 16 000: M1 only if the first step is shown correctly. (b) B1: to avoid bias; so that she does not choose places with many (or few) dandelions; so that the sample is representative of the whole field; so the estimate is fair or valid. "To make it accurate" alone scores 0; "to make it a fair test" scores 0 (this is not a fair test).	Quadrat sampling is 高二 (选择性必修2, population density by 样方法). The calculation is accessible to any student who can find a mean and scale up by area: do not mark NT unless the student says so. The word "quadrat" may be unknown: the diagram and the printed area carry the meaning.

Q	Topic	Answer	Marks and partial credit	May not have been taught?
7	Natural selection (explain, 2 marks)	Dark beetles are better camouflaged on dark soil, so fewer are eaten by birds; they survive and reproduce, passing the allele (gene) for dark colour to their offspring, so the proportion of dark beetles rises each generation	B1 for the **selection** idea: dark beetles are camouflaged (harder for birds to see), so more of them survive **or** more light beetles are eaten. B1 for the **inheritance** idea: the surviving dark beetles reproduce and pass on the allele (gene) for dark colour, so more of the next generation are dark. Do not accept "the beetles changed colour to match the soil" or "the beetles needed to be dark" (Lamarckian or purposive explanations): 0 for that statement, but the other mark can still be earned. A bare description of the chart ("more beetles became dark") scores 0.	Natural selection is 初二 (生物的进化) and the end of 必修2: usually taught in content. Writing a two-step explanation in English is the challenge: an answer that gives both ideas in simple English scores 2.

Band guide (provisional, Section 7): 0 to 4: **1**; 5 to 7: **2**; 8 to 11: **3**; 12 to 15: **4**. Look at the NT and L questions before deciding (Section 7).

Not tested (not enough time): respiration, transport in plants and humans, gas exchange, the nervous system, reproduction, DNA structure, food chains and energy flow. Ask about these in the interview if they matter for the student's subject choices.

What to look for

- **Explaining in English** (questions 3b and 7) is the best guide to readiness for A Level biology, which asks for long written answers using precise terms (osmosis, water potential, allele, selection). A student who gives the right ideas in plain English is ready for the content and needs the vocabulary; a student who describes the data instead of explaining it needs teaching in how to answer "explain" questions.
- **Working scientifically** (question 6): a student who scales up correctly and can say why the sample is random has the skills for A Level practical work. Watch for the estimate given without working: ask them in the interview how they did it.
- **An English barrier** looks like blank or wrong answers that become right when you read the question aloud, or answers in the wrong form (a description where an explanation was asked). The words most likely to be unknown: denatured, optimum, genotype, offspring, quadrat, random, population.

5. Chemistry (15 minutes, 15 marks)

What a Year 12 entrant may not have been taught. Chinese 高一 chemistry (必修1 and 必修2) covers atomic structure, the periodic table, ionic and covalent bonding, **the mole and amount of substance**, redox and ionic equations, a first look at reaction rates and reaction energy, and a short introduction to organic compounds (methane, ethene, ethanol). Electrolysis (beyond the electrolysis of water in 初三), titration, equilibrium, collision theory and organic chemistry in depth are **高二** (选择性必修). So a Year 12 entrant usually has the moles calculation (question 4) but may mark NT on questions 3 (homologous series), 5 (electrolysis products), 6 (activation energy) and 8 (titration). The titration calculation itself is a moles calculation and is often within reach.

Q	Topic	Answer	Marks and partial credit	May not have been taught?
1	Atomic structure	12	B1 (23 – 11). An answer of 11 (protons or electrons) or 23 (mass number) scores 0.	Secure (初三 and 高一). Chinese periodic tables print the atomic number **above** the symbol and the relative atomic mass below: the diagram labels both numbers, so a wrong reading is a science error, not a format error.

Q	Topic	Answer	Marks and partial credit	May not have been taught?
2	Bonding and structure: data and explain	(a) X (b) When molten the ions are free to move (and carry the charge); when solid the ions are held in fixed positions in the lattice	(a) B1. (b) B1: both halves needed in some form: ions **can move** when molten **and** cannot move (are fixed) when solid. "Electrons are free to move" scores 0 (it is ions that move in a molten ionic compound).	Ionic and covalent bonding and the conduction of molten or dissolved electrolytes are 高一 必修1: usually taught.
3	Organic chemistry: homologous series	alkenes	B1. Accept "alkene". "Alkanes" scores 0; "hydrocarbons" alone scores 0 (the question says it is a hydrocarbon); "unsaturated hydrocarbons" scores 0 (not a homologous series).	Ethene is introduced in 必修2 (end of 高一); homologous series (同系物) and the alkene family are usually **高二**: may not have been taught. The English term "homologous series" may be unknown even when the idea is: if the student asks what it means, do not explain; mark L only if a colleague translates.
4	Reacting masses (moles)	14.0 g	M1 for Mr of $CaCO_3$ = 100 **or** moles of $CaCO_3$ = 25.0 ÷ 100 = 0.250 mol; M1 for moles of CaO = 0.250 (1 : 1 ratio) **or** Mr of CaO = 56 used with the ratio; A1 for 14.0 g (accept 14). The ratio method 25.0 × 56 ÷ 100 = 14.0 scores all three. Full marks for 14.0 with no working. Common wrong answers: 11.0 g (mass of CO_2: M1 M1 A0, as the method is correct but the wrong product is found); 25.0 × 56 = 1 400 (M1 at most).	Moles and reacting masses are 高一 必修1: usually taught. The strongest guide to A Level chemistry readiness on this paper.
5	Electrolysis of a molten salt	lead	B1. Accept Pb. "Lead ions" scores 0; "bromine" scores 0 (that is the positive electrode); "hydrogen" scores 0 (there is no water: the salt is molten).	Electrolysis of molten salts is 高二 (选择性必修1): may not have been taught. A student who reasons from charges (positive lead ions go to the negative electrode) can still answer.
6	Energetics: reaction profile	A	B1.	Activation energy appears in 必修2 (高一) briefly and in 高二 in depth: may not have been taught.

Q	Topic	Answer	Marks and partial credit	May not have been taught?
7	Rates of reaction: data and explain (2 marks)	(a) 100 s (b) At the higher temperature the particles have more (kinetic) energy and move faster, so they collide more often; more of the collisions have enough energy (the activation energy) to react, so more collisions are successful	(a) B1: 100 s (accept 95 to 105). (b) B1 for particles having more energy / moving faster **so** collisions are more frequent; B1 for a greater proportion of collisions having energy equal to or above the activation energy (more successful collisions). "The particles move faster" alone scores 1 only if linked to more collisions; "the particles collide more" alone scores 0 without the energy or speed idea. "More particles" or "higher concentration" scores 0 (the temperature changed, not the amount).	Factors affecting rate are 必修2 (高一); collision theory with activation energy is 高二: the first mark is usually within reach, the second may not have been taught. A student who gives only the first idea in good English is at IGCSE level; both ideas is A Level ready.
8	Titration: working scientifically and calculation	(a) 1 and 2 (b) 0.0800 mol/dm^3	(a) B1: titrations 1 and 2 (the two concordant results, within 0.10 cm^3 of each other; the rough titration and titration 3 are not used). (b) M1 for moles of HCl = 0.100 × 20.00 ÷ 1000 = 0.002 00 mol (or 2.00 × 10^{-3}); A1 for concentration = 0.002 00 ÷ 0.0250 = 0.0800 mol/dm^3 (accept 0.08). Full marks for 0.08 with no working. A student who uses a different mean titre (for example 20.25 from all four) can score M1 only, since the question gives 20.00. Common wrong answer: 0.125 (concentration × 25 ÷ 20, inverted): M0 A0 unless moles are correctly found first.	Titration is 高二 (选择性必修1): the idea of concordant titres may not have been taught, and (a) is often NT or wrong. The calculation is moles (高一): many students can do (b) from the equation and the printed data. Judge the two parts separately.

Band guide (provisional, Section 7): 0 to 4: **1**; 5 to 7: **2**; 8 to 11: **3**; 12 to 15: **4**. Look at the NT and L questions before deciding (Section 7).

Not tested (not enough time): the periodic table groups and trends, acids, bases and salts, equilibrium, metals and the reactivity series, tests for ions and gases, the chemistry of alcohols and polymers. Ask about these in the interview if they matter for subject choices.

What to look for

- **The moles calculations** (questions 4 and 8b) are the best single guide to readiness for A Level chemistry, which uses them from the first weeks. Clear working (moles, ratio, mass or concentration) with units is a strong response; a correct answer with no working is fine but tells you less.
- **The explain questions** (2b and 7b) show whether the student can argue in terms of particles in English. Watch for ions and electrons confused in 2b, and for "more particles" in 7b (a change of concentration, not temperature): these are science errors, not English ones.
- **An English barrier** looks like: the calculation done correctly but (a) parts blank, or correct ideas written as single words without the connecting "so" or "because". Words most likely to be unknown: neutron (usually known), molten, homologous series, decomposes, electrode, activation energy, concordant (not used on the paper), titration.

6. Physics (15 minutes, 15 marks)

What a Year 12 entrant may not have been taught. Chinese 高一 physics (必修1 and 必修2) covers motion and graphs, forces and Newton's laws, work, energy and power, and circular motion and gravitation; **momentum** is at the end of 高一 or the start of 高二, depending on the textbook edition (选择性必修1 in the 2019 人教版). Circuits and Ohm's law are 初三 (and again in 高二 必修3); electromagnetic induction and transformers are **高二** (选择性必修2); radioactivity and half-life are **高二 or 高三** (选择性必修3). So questions 1 to 4 are usually taught; expect NT most on question 5b (why a transformer needs a.c.), question 6 (half-life and background radiation) and sometimes question 7 (momentum).

Q	Topic	Answer	Marks and partial credit	May not have been taught?
1	Velocity-time graph: acceleration (data)	3 m/s²	B1 (12 ÷ 4). "12" (the velocity) or "24" (the distance) scores 0.	Secure (高一 必修1).
2	Resultant force and F = ma	(a) 30 N (b) 3 m/s²	(a) B1: 30 (to the right; direction not required). (b) B1: 3; ecf: their (a) ÷ 10, so 5 m/s² from a resultant of 50 N scores (b). An answer of 7 (70 ÷ 10) scores (a) 0 and (b) 1 by ecf only if 70 is written in (a).	Secure (高一 必修1, Newton's second law).
3	Gravitational potential energy and power	(a) 3 000 J (b) 300 W	(a) B1: 60 × 10 × 5 = 3 000. (b) B1: 3 000 ÷ 10 = 300; ecf their (a) ÷ 10. Using g = 9.8 gives 2 940 J and 294 W: accept both (the paper says g = 10 N/kg, but 9.8 is what Chinese schools use).	Secure (初二 and 高一 必修2).
4	Series circuit: resistance and current	(a) 6 Ω (b) 2 A	(a) B1. (b) B1: 12 ÷ 6 = 2; ecf 12 ÷ their (a) (so 3 A from 4 Ω scores). An answer of 2 A with (a) blank scores 1 for (b) only.	Usually secure (初三 Ohm's law and series circuits; 高二 in depth).
5	Transformer: calculation and explain (2 marks)	(a) 12 V (b) A voltage is induced in the secondary coil only when the magnetic field in the core is **changing**; a.c. in the primary gives a changing field, but d.c. gives a steady (constant) field, so no voltage is induced	(a) B1: 220 × 60 ÷ 1100 = 12. "4 033 V" (inverted ratio) scores 0. (b) B1 for d.c. producing a steady or constant (unchanging) magnetic field in the core; B1 for the principle: a voltage (current) is induced in the secondary only when the magnetic field (flux) through it changes. "There is no current in the secondary" alone scores 0 (restates the question). "d.c. does not have a frequency" alone scores 0.	Transformers and induction are 高二 (选择性必修2): may not have been taught; 初三 touches on generators. The calculation uses the printed equation and is often within reach even when NT. (b) is the hardest explain question on the paper: a correct answer is strong evidence of A Level readiness.
6	Radioactivity: half-life (data) and working scientifically	(a) 2 days (b) To measure the background count rate, which must be subtracted from the readings (so that the readings show the source alone)	(a) B1: 2 days (800 to 400, or any halving). (b) B1: background radiation (from rocks, cosmic rays, the air, the building) is always present; measuring it lets him subtract it from the readings (or correct them), so that the readings show the source alone. "To check the detector works" scores 0.	Radioactivity is 高二 or 高三 (选择性必修3): may not have been taught. The half-life reading can be made by any student who understands "half": do not mark NT unless the student says so; (b) is more often NT.
7	Momentum and conservation	(a) 6 kg m/s (b) 2 m/s	(a) B1: 2 × 3 = 6. (b) M1 for conservation of momentum: (2 + 1) × v = 6, or momentum after = 6 (ecf their (a)) with total mass 3 kg; A1 for 2 m/s. Full marks for 2 m/s with no working. Common wrong answers: 3 m/s (velocity unchanged: 0); 6 m/s (momentum given as a velocity: 0); 1.5 m/s (kinetic energy shared or 6 ÷ 4: M0 A0).	Momentum is the end of 高一 or 高二 depending on the edition: may not have been taught. A student who has met it usually finds this straightforward: the question is here because A Level physics assumes it early.

Band guide (provisional, Section 7): 0 to 4: **1**; 5 to 7: **2**; 8 to 11: **3**; 12 to 15: **4**. Look at the NT and L questions before deciding (Section 7).

Not tested (not enough time): density and pressure, moments, thermal physics, waves and the electromagnetic spectrum, parallel circuits and electrical power, the motor effect, types of radiation and their penetration. Ask about these in the interview if they matter for subject choices.

What to look for

- **Choosing the right equation** from the box, substituting with units and giving the answer with a sensible unit (questions 2, 3, 4, 7) is what A Level physics needs on day one. A student who cannot choose the equation when it is printed in front of them is not yet ready, even if their arithmetic is right.
- **The two-step problems** (questions 2b, 3b, 7b) separate a strong IGCSE student from a weak one: look for the first result used correctly in the second step, and give ecf as the table says.
- **An English barrier** looks like: the calculations right but the explain and working-scientifically parts (5b, 6b) blank, or correct physics in single words ("change", "field") without a sentence. Words most likely to be unknown: resultant, staircase, induced, background radiation, trolley, stick together.

7. Band guide and recording

Band guide for each science (provisional, the same for all three):

Marks out of 15	Band	What it usually means
0 to 4	**1**	Well below the level needed to start an A Level in this science.
5 to 7	**2**	Working towards: some IGCSE content secure, gaps in the rest or in explaining and working scientifically.
8 to 11	**3**	Ready for A Level in this science, usually with some topics to catch up.
12 to 15	**4**	Strong: ready, with the stretch questions answered.

Before you give the band, look at NT and L. An NT question scores 0 but tells you about the student's school sequence, not their ability: a Grade 10 student may reasonably mark NT on electrolysis, titration, transformers, radioactivity or quadrats. Judge whether the student would reach the next band once the NT topics are taught, and say so in the comment. A high number of L questions (two or more in one science) means the English of the paper, not the science, limited the result: say so, and look at the English bands together with the science. A band 1 that NT or L explains is treated as band 2 for the readiness judgement, as in the rest of the pack.

Recording. On the grading sheet (`Y12_Grading_Sheet_v3.1`), for each optional science: the mark / 15, the band, the NT questions (by number and topic), the number of L questions, or a tick in "not taken". Note in the comment which questions were left blank at the end of the 15 minutes (time) and anything you observed (for example calculator use, working shown, confidence). If a science was given on a second visit, write the date.

8. How the result feeds the readiness judgement for A Level sciences

This section is guidance for the school to confirm against its own entry requirements for each A Level science and the support it can give. It is not a set of fixed rules. Read each science result with the English bands, the maths result and the core science mark (Section 6 of `Y12_Assessor_Instructions_v3.1`, which has the full readiness table for the year).

English is the gateway. A Level sciences are taught and examined in English, with long explain and evaluate answers, practical write-ups and technical vocabulary. As a working guide: **English band C or above (reading and writing) with ESL support** is the minimum to start an A Level science taught in English; **band D or above** to follow it with little extra support. A student with strong science bands and English at band B is not yet ready for A Level science in English, however good the science.

Maths matters for chemistry and physics. A Level physics and chemistry assume fluent algebra, rearranging equations, ratios, standard form and graphs. Read the physics and chemistry results with the maths result: maths band 3 or 4 supports them; maths band 1 or 2 does not, whatever the science band.

Science result (allowing for NT and L)	What it suggests for that A Level science (for the school to confirm)
Band 4	**Ready.** Likely to do well; consider it for the strongest subject choices.
Band 3	**Ready**, with the NT topics to catch up in the first term (list them on the grading sheet and pass them to the subject teacher).

Science result (allowing for NT and L)	What it suggests for that A Level science (for the school to confirm)
Band 2, mostly explained by NT (the topics that are 高二 in China)	**Probably ready with a learning plan**: the taught topics are secure. Set a short catch-up plan for the NT topics and review after six weeks.
Band 2 with errors on taught topics (for example biology 1 to 3, chemistry 1, 2 and 4, physics 1 to 4)	**Not yet ready for this science at A Level.** Discuss a one-year pre-A Level or IGCSE route, or a different subject choice, with the family.
Band 1 not explained by NT or English	**Not ready.** Do not offer this science at A Level; if the student insists, a one-year IGCSE or pre-A Level route is the realistic step.
Band 3 or 4 in science, English band C	Ready in science; **ESL support** and a limit on the number of essay-based subjects. Two sciences plus maths is often a better load than three sciences while English grows.
Band 3 or 4 in science, English band B	**Not yet ready for A Level sciences taught in English.** A one-year pre-A Level or IGCSE route with intensive English is the usual advice (`00_Admissions_Guide.md`, Section 9).
Physics or chemistry band 3 or 4, maths band 1 or 2	Check the maths NT first. If the maths is genuinely weak, A Level physics or chemistry will be hard: offer with maths support, or advise a different combination.
High marks on calculations, low marks on explain and working-scientifically questions in every science	Strong content, less practice in the written and practical sides of science in English. Ready, with teaching of "explain" answers and practical write-ups from the start.

Reading all three together. A student ready in all the sciences they chose, with English D or E and maths 3 or 4, is ready for a full A Level programme. A student ready in one science but not another should be advised on subject choices before an offer. Use the Year 12 interview questions (subject choices and why, university plans, what they know about A Level) to check that the choices are realistic and the student's own. Record the readiness judgement for each science on the grading sheet and on the final mark sheet (`forms/Candidate_Summary_and_Decision`).